Lawrence Du, Ph.D.

Machine Learning Engineering • Computational Biology • MLOps
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Skills

Languages: Python, SQL, C/C++, C#, Rust, Java, Dart, Bash, HTML/CSS, MATLAB
ML & data: PyTorch, TensorFlow, scikit-learn (sklearn), NumPy, pandas, Numba, Ray, MLflow, Flask
Cloud/MLOps: Amazon Web Services (AWS), Google Cloud Platform (GCP), Kubernetes, Docker, Terraform, Pulumi, Metaflow
ML & scientific computing: Foundation model fine-tuning, deep learning, transformers, convolutional neural networks (CNNs), graph neural networks (GNNs), diffusion, Louvain/Leiden community detection, SVM, PCA, KNN, decision trees

Experience

GSK • **Senior AI/ML Engineer, Sequence Learning and Phenomics**
Oct 2024 – Present | South San Francisco, CA

- Built a sequence foundation model fine-tuning platform for RNA properties and single-cell RNA-seq perturbation prediction; cut iteration time **from weeks to less than a day**.
- Integrated Segment Anything Model (SAM) image segmentation into production computer vision and embedding pipelines for single-cell tracking and cell-state world modeling across **millions of images**.
- Designed embedding-based differentiation QC for **>3 million iPSC-to-microglia cells** in Alzheimer's research.
- Piloted generative AI for protein subcellular localization images: x-prediction / JiT diffusion using ESM protein embeddings.
- Built a pipeline using large language models (LLMs) for retrieval-augmented generation (RAG) and gene-candidate evaluation; prioritized **~1,000 genes** for a lung-epithelium CRISPR knockout screen supporting a COPD drug program.
- Shipped Model Context Protocol (MCP) servers enabling company-wide LLM assistants to build and launch image-analysis pipelines; accelerated analysis turnaround **>10×**.
- Conducted interviews for team hiring; contributing to LLM benchmarking currently in development.

Freenome • **Senior Machine Learning Research Engineer**
Aug 2022 – Jun 2024 | South San Francisco, CA

- Led greenfield distributed ML platform development with PyTorch, Ray, and Kubernetes for DNA methylation/protein cancer detection; accelerated model training **>10×** with distributed data parallelism (DDP).
- Deployed organization-wide MLflow experiment tracking and automated reporting on GCP with Terraform/Pulumi; reduced manual research effort **≥5×**.
- Built PyTorch multitask neural networks and elastic net models for colorectal cancer risk from cell-free DNA in **>27,000 clinical trial participants**.
- Piloted GPT-4 biomedical natural language processing (NLP) and fine-tuned an open-source LLM with direct preference optimization (DPO) for literature summarization and feature extraction.

23andMe • **Machine Learning Engineer / Data Scientist**
Nov 2018 – Aug 2022 | Sunnyvale, CA | Data Scientist prior to 2020

- Deployed Recent Ancestor Locations (RAL) country matching for **>15 million customers**; used identity-by-descent (IBD) graphs and community detection to identify genetic subpopulations.
- Built large-scale data pipelines for SNP feature engineering and ETL (**~10 million samples × ~1 million SNPs**) with AWS Batch/Glue/Athena and Metaflow; enabled genome-wide association studies (GWAS) and polygenic risk scores (PRS).
- Improved type 2 diabetes and coronary artery disease PRS with stacking ensembles; automated MLflow/Jupyter model evaluation, cutting analysis **from days to minutes**.

Scripps Research • **Bioinformatician IV** • **May 2018 – Oct 2018**
Built an RNA-based transplant rejection classifier and Nanopore bioinformatics pipelines with Common Workflow Language (CWL).

Juno Diagnostics • **Independent Consultant** • **Sep 2017 – Feb 2018**
Developed a TensorFlow prenatal screening classifier and co-invented US20210020314A1: Deep learning-based prenatal testing.

Insight • **Data Science Fellow** • **Jan 2017 – Apr 2017**
Built DeepPixelMonster, a TensorFlow generative adversarial network (GAN) for pixel art; deployed with Flask on AWS EC2.

Education

Ph.D., Biology • UC San Diego • 2010–2017 • Scott A. Rifkin Lab
Research: DeepNuc CNNs for **>500,000** transcription start site and **>60,000** microRNA target sequences; MATLAB image segmentation, fluorescence quantification, and deconvolution of RNA expression in **>5,000 embryos**.
B.A., Biological Sciences • Cornell University • 2006–2010 • Genetics and Development, *magna cum laude*

Selected Projects

Rogue Stargun — Unity/C# VR space simulator; behavior trees, AI flight planning, procedural missions, text-to-speech.
DiffuMon — Open-source denoising diffusion probabilistic model (DDPM) image generator.